

3.7 Systems of Linear Equations [Gauss Seidel – Indirect Method]

Direct methods for solving systems of linear equations:

- Gaussian elimination
- Other methods (LU decomposition, matrix inversion)

Indirect (iterative) methods

- Gauss Seidel – employs initial guesses, and then we iterate to converge on a solution.
- This formulation is well-suited to large numbers of equations and doesn't induce the round-off error seen in Gaussian elimination.
 - In iterative methods, round-off errors do not accumulate (as they do in Gaussian elimination) since each iteration is essentially a new approximation to the solution.
 - We do have round-off in Gauss-Seidel, but it is only associated with the current iteration (whereas in Gaussian elimination, error can accumulate for the sequence of steps in the solution)
- The method has improved efficiency, doesn't store or make calculations using zero values.

3.7 Systems of Linear Equations [Gauss Seidel]

- Express equations in an explicit form (isolate each variable).
- Let's look at a 3x3 system.

In matrix form
 $[A] \{x\} = \{b\}$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

$$\text{Row\#1 } x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3]$$

$$\text{Row\#2 } x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3]$$

$$\text{Row\#3 } x_3 = \frac{1}{a_{33}} [b_3 - a_{31}x_1 - a_{32}x_2]$$

Note:

- Diagonal elements $\neq 0$ (divide by zero).
- Convergence is not guaranteed.

3.7 Systems of Linear Equations [Gauss Seidel]

In Gauss Seidel, we solve the equations using an iterative approach:

- Step 1: 'Guess' initial values for the unknowns (x_i):
$$\begin{cases} x_1 = 0.0 \\ x_2 = 0.0 \\ x_3 = 0.0 \end{cases}$$
- Step 2: Solve for new x values (e.g. x_1 at iteration $i+1$) using the Row#1 equation, solved in terms of x_1 .
- Step 3: Solve for new x_n values (x_1) using values from the previous steps.
- Step 4: Repeat until the convergence criterion is met.

$x_{\text{new}} - x_{\text{old}} / x_{\text{new}}$

$$|e_{a,i}| = \left| \frac{x_i^j - x_i^{j-1}}{x_i^j} \right| 100\% < \epsilon_s$$

Each individual relative approximate error ($\epsilon_{a,i}$) should be less than the convergence criterion (ϵ_s)

Alternative (more conservative), the sum of the approximate errors must be less than the convergence criterion.

3.7 Systems of Linear Equations [Gauss Seidel]

Example 3.7a [3x3 system of equations]:

$$2x_1 + x_2 + x_3 = 3$$

$$x_1 + 2x_2 - x_3 = 6$$

$$x_1 - x_2 + 3x_3 = -4$$

Rearrange equations:

$$x_1 = \frac{1}{2}[3 - x_2 - x_3]$$

$$x_2 = \frac{1}{2}[6 - x_1 + x_3]$$

$$x_3 = \frac{1}{3}[-4 - x_1 + x_2]$$

Guess
initial
values

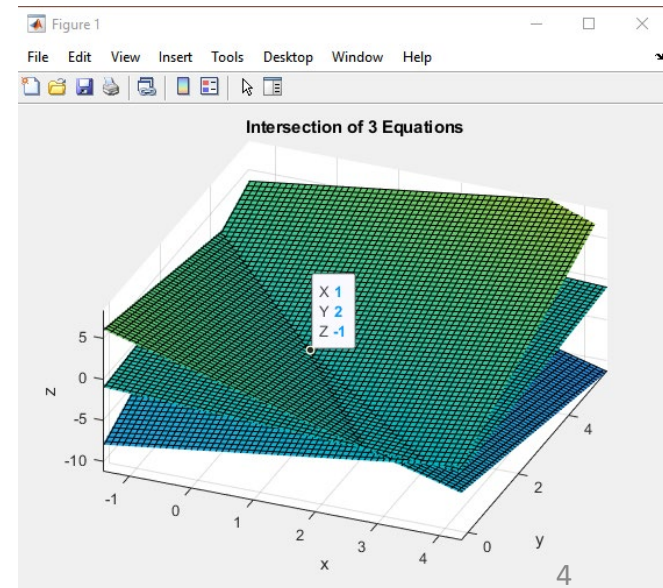
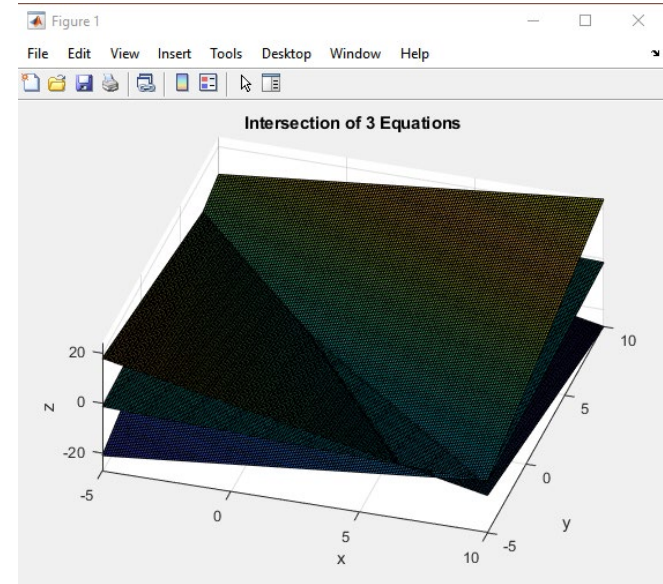
Iteration #

$$x_1^0 = 1.0$$

$$x_2^0 = 1.0$$

$$x_3^0 = 1.0$$

Graphical Method



3.7 Systems of Linear Equations [Gauss Seidel]

$$x_1^0 = 1.0$$
$$x_2^0 = 1.0$$
$$x_3^0 = 1.0$$

$$x_1^1 = \frac{1}{2} [3 - x_2^0 - x_3^0] \qquad x_2^1 = \frac{1}{2} [6 - x_1^1 + x_3^0] \qquad x_3^1 = \frac{1}{3} [-4 - x_1^1 + x_2^1]$$

					Convergence ->		1.00E-06
Iteration #	x1	x2	x3		εa1	εa2	εa3
0	1.000000000000	1.000000000000	1.000000000000				
1	0.500000000000	3.250000000000	-0.416666666667	-	1.0000E+02	6.9231E+01	3.4000E+02
2	0.083333333333	2.750000000000	-0.444444444444	-	5.0000E+02	1.8182E+01	6.2500E+00
3	0.347222222222	2.604166666667	-0.58101851852	-	7.6000E+01	5.6000E+00	2.3506E+01
4	0.48842592593	2.465277777778	-0.67438271605	-	2.8910E+01	5.6338E+00	1.3844E+01
5	0.60455246914	2.36053240741	-0.74800668724	-	1.9209E+01	4.4374E+00	9.8427E+00
6	0.69373713992	2.27912808642	-0.80486968450	-	1.2856E+01	3.5717E+00	7.0649E+00
69	0.99999996930	2.00000002798	-0.99999998044	Converged	8.948E-07	4.078E-07	5.701E-07



				Convergence->			1.00E-06
Iteration #	x1	x2	x3		εa1	εa2	εa3
0	1.000000000000	1.000000000000	1.000000000000				
1	0.500000000000	3.250000000000	-0.416666666667	-	1.000E+02	6.923E+01	3.400E+02
2	0.083333333333	2.750000000000	-0.444444444444	-	5.000E+02	1.818E+01	6.250E+00
3	0.347222222222	2.604166666667	-0.58101851852	-	7.600E+01	5.600E+00	2.351E+01
4	0.48842592593	2.465277777778	-0.67438271605	-	2.891E+01	5.634E+00	1.384E+01
5	0.60455246914	2.36053240741	-0.74800668724	-	1.921E+01	4.437E+00	9.843E+00
6	0.69373713992	2.27912808642	-0.80486968450	-	1.286E+01	3.572E+00	7.065E+00
7	0.76287079904	2.21612975823	-0.84891368027	-	9.062E+00	2.843E+00	5.188E+00
8	0.81639196102	2.16734717936	-0.88301492722	-	6.556E+00	2.251E+00	3.862E+00
9	0.85783387393	2.12957559942	-0.90941942484	-	4.831E+00	1.774E+00	2.903E+00
10	0.88992191271	2.10032933123	-0.92986419383	-	3.606E+00	1.392E+00	2.199E+00
11	0.91476743130	2.07768418744	-0.94569441462	-	2.716E+00	1.090E+00	1.674E+00
12	0.93400511359	2.06015023589	-0.95795162590	-	2.060E+00	8.511E-01	1.280E+00
13	0.94890069500	2.04657383955	-0.96744228515	-	1.570E+00	6.634E-01	9.810E-01
14	0.96043422280	2.03606174602	-0.97479082559	-	1.201E+00	5.163E-01	7.539E-01
15	0.96936453978	2.02792231731	-0.98048074082	-	9.213E-01	4.014E-01	5.803E-01
16	0.97627921176	2.02162002371	-0.98488639602	-	7.083E-01	3.117E-01	4.473E-01
17	0.98163318615	2.01674020892	-0.98829765908	-	5.454E-01	2.420E-01	3.452E-01
18	0.98577872508	2.01296180792	-0.99093897239	-	4.205E-01	1.877E-01	2.665E-01
19	0.98898858223	2.01003622269	-0.99298411985	-	3.246E-01	1.455E-01	2.060E-01
20	0.99147394858	2.00777096579	-0.99456766093	-	2.507E-01	1.128E-01	1.592E-01
21	0.99339834757	2.00601699575	-0.99579378394	-	1.937E-01	8.744E-02	1.231E-01
22	0.99488839410	2.00465891098	-0.99674316104	-	1.498E-01	6.775E-02	9.525E-02
23	0.99604212503	2.00360735697	-0.99747825602	-	1.158E-01	5.248E-02	7.370E-02
24	0.99693544953	2.00279314723	-0.99804743410	-	8.961E-02	4.065E-02	5.703E-02
25	0.99762714344	2.00216271123	-0.99848814407	-	6.933E-02	3.149E-02	4.414E-02
26	0.99816271642	2.00167456976	-0.99882938222	-	5.366E-02	2.439E-02	3.416E-02
27	0.99857740623	2.00129660577	-0.99909360015	-	4.153E-02	1.889E-02	2.645E-02
28	0.99889849719	2.00100395133	-0.99929818195	-	3.214E-02	1.463E-02	2.047E-02
29	0.99914711531	2.00077735137	-0.99945658798	-	2.488E-02	1.133E-02	1.585E-02
30	0.99933961831	2.00060189686	-0.99957924048	-	1.926E-02	8.770E-03	1.227E-02
31	0.99948867181	2.00046604385	-0.99967420932	-	1.491E-02	6.791E-03	9.500E-03
32	0.99960408274	2.00036085397	-0.99974774292	-	1.155E-02	5.259E-03	7.355E-03
33	0.99969344447	2.00027940630	-0.99980467939	-	8.939E-03	4.072E-03	5.695E-03
34	0.99976263654	2.00021634203	-0.99984876484	-	6.921E-03	3.153E-03	4.409E-03
35	0.99981621140	2.00016751188	-0.99988289984	-	5.358E-03	2.441E-03	3.414E-03
36	0.99985769398	2.00012970309	-0.99990933030	-	4.149E-03	1.890E-03	2.643E-03
37	0.99988981360	2.00010042805	-0.99992979518	-	3.212E-03	1.464E-03	2.047E-03
38	0.99991468357	2.00007776062	-0.99994564098	-	2.487E-03	1.133E-03	1.585E-03
39	0.99993394018	2.00006020942	-0.99995791025	-	1.926E-03	8.775E-04	1.227E-03
40	0.99994885042	2.00004661967	-0.99996741025	-	1.491E-03	6.795E-04	9.500E-04
41	0.99996039529	2.00003609723	-0.99997476602	-	1.155E-03	5.261E-04	7.356E-04
42	0.99996933440	2.00002794979	-0.99998046154	-	8.939E-04	4.074E-04	5.696E-04
43	0.99997625587	2.00002164130	-0.99998487153	-	6.922E-04	3.154E-04	4.410E-04
44	0.99998161511	2.00001675668	-0.99998828614	-	5.359E-04	2.442E-04	3.415E-04
45	0.99998576473	2.00001297456	-0.99999093006	-	4.150E-04	1.891E-04	2.644E-04
46	0.99998897775	2.00001004610	-0.99999297722	-	3.213E-04	1.464E-04	2.047E-04
47	0.99999146556	2.00000777861	-0.99999456232	-	2.488E-04	1.134E-04	1.585E-04
48	0.99999339185	2.00000602292	-0.99999578965	-	1.926E-04	8.778E-05	1.227E-04
49	0.99999488336	2.00000466350	-0.99999673996	-	1.492E-04	6.797E-05	9.503E-05
50	0.99999603823	2.00000361091	-0.99999747577	-	1.155E-04	5.263E-05	7.358E-05
51	0.99999693243	2.00000279590	-0.99999804551	-	8.942E-05	4.075E-05	5.697E-05
52	0.99999762481	2.00000216484	-0.99999848666	-	6.924E-05	3.155E-05	4.411E-05
53	0.99999816091	2.00000167622	-0.99999882823	-	5.361E-05	2.443E-05	3.416E-05
54	0.99999857601	2.00000129788	-0.99999909271	-	4.151E-05	1.892E-05	2.645E-05
55	0.99999889741	2.00000100494	-0.99999929749	-	3.214E-05	1.465E-05	2.048E-05
56	0.99999914628	2.00000077812	-0.99999945605	-	2.489E-05	1.134E-05	1.586E-05
57	0.99999933897	2.00000060249	-0.99999957883	-	1.927E-05	8.781E-06	1.228E-05
58	0.99999944817	2.00000046650	-0.99999967389	-	1.492E-05	6.799E-06	9.506E-06
59	0.99999960369	2.00000036121	-0.99999974749	-	1.155E-05	5.265E-06	7.361E-06
60	0.99999969314	2.00000027968	-0.99999980449	-	8.945E-06	4.076E-06	5.699E-06
61	0.99999976240	2.00000021655	-0.99999984862	-	6.926E-06	3.156E-06	4.413E-06
62	0.99999981603	2.00000016768	-0.99999988278	-	5.363E-06	2.444E-06	3.417E-06
63	0.99999985755	2.00000012983	-0.99999990924	-	4.152E-06	1.892E-06	2.646E-06
64	0.99999988971	2.00000010053	-0.99999992973	-	3.215E-06	1.465E-06	2.049E-06
65	0.99999991460	2.00000007784	-0.99999994559	-	2.489E-06	1.134E-06	1.586E-06
66	0.99999993388	2.00000006027	-0.99999995787	-	1.928E-06	8.784E-07	1.228E-06
67	0.99999994880	2.00000004667	-0.99999996738	-	1.492E-06	6.802E-07	9.509E-07
68	0.99999996036	2.00000003613	-0.99999997474	-	1.156E-06	5.266E-07	7.363E-07
69	0.99999996930	2.00000002798	-0.99999998044	Converged	8.948E-07	4.078E-07	5.701E-07

In this case, each individual relative error is less than the convergence criterion (1x10⁻⁶).

3.7 Systems of Linear Equations [Gauss Seidel]

Convergence: Check using approximate relative error.

$$\epsilon_{an} = \left| \frac{x_n^i - x_n^{i-1}}{x_n^i} \right| \times 100\%$$

for all n values of x

Notes:

- Diagonal terms must not equal zero to avoid division by zero.
 - *Apply pivoting if required to move equations and avoid division by zero.
- Convergence is not guaranteed in all cases.

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

Gauss Seidel is an indirect (iterative) method for solving systems of linear equations.

- Algorithm is easy to implement.
- Efficient, particularly for banded matrices.
- Can be slow to converge and can diverge.

Two approaches to verify convergence and improve the convergence characteristics of the Gauss Seidel solution.

1. Convergence criterion (diagonal dominant).
2. Relaxation (changes the convergence rate).

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

1. Convergence criterion (diagonal dominant)

For Gauss Seidel, the solution of $[A]\{x\}=\{b\}$, convergence will occur only if $[A]$ is diagonally dominant. [section 11.2 in Chapra]

$$|a_{ii}| > \sum_{j=1; j \neq i}^n |a_{ij}|$$

This criterion is sufficient but not necessary for convergence. If the condition is satisfied, convergence is guaranteed, but if the condition is not met, convergence may still be achieved.

Systems of equations where this holds are called diagonally dominant problems. Many practical problems in engineering are diagonally dominant.

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

1. Convergence criterion (diagonal dominant)

Example 3.8a

$$2x_1 + x_2 + x_3 = 3$$

$$x_1 + 2x_2 - x_3 = 6$$

$$x_1 - x_2 + 3x_3 = -4$$

$$[A] = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

$$|a_{ii}| > \sum_{j=1; j \neq i}^n |a_{ij}|$$

Row#1

$$|a_{11}| = |a_{12}| + |a_{13}|$$
$$2 = 1 + 1$$

Row#2

$$2 = 1 + 1$$

Row#3

$$3 > 1 + 1$$

Therefore, convergence is possible, but not guaranteed.

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

1. Convergence criterion (diagonal dominant)

Example 3.8a (as before and shown in ex. 3.7a)

$$2x_1 + x_2 + x_3 = 3$$

$$x_1 + 2x_2 - x_3 = 6$$

$$x_1 - x_2 + 3x_3 = -4$$

Iteration #	x1	x2	x3		Convergence ->		
					εa1	εa2	εa3
0	1.000000	1.000000	1.000000				1.00E-06
1	0.500000	3.250000	-0.416667	-	1.00E+02	6.92E+01	3.40E+02
2	0.083333	2.750000	-0.444444	-	5.00E+02	1.82E+01	6.25E+00
3	0.347222	2.604167	-0.581019	-	7.60E+01	5.60E+00	2.35E+01
4	0.488426	2.465278	-0.674383	-	2.89E+01	5.63E+00	1.38E+01
5	0.604552	2.360532	-0.748007	-	1.92E+01	4.44E+00	9.84E+00
6	0.693737	2.279128	-0.804870	-	1.29E+01	3.57E+00	7.06E+00
7	0.762871	2.216130	-0.848914	-	9.06E+00	2.84E+00	5.19E+00
8	0.816392	2.167347	-0.883015	-	6.56E+00	2.25E+00	3.86E+00
9	0.857834	2.129576	-0.909419	-	4.83E+00	1.77E+00	2.90E+00
10	0.889922	2.100329	-0.929864	-	3.61E+00	1.39E+00	2.20E+00
11	0.914767	2.077684	-0.945694	-	2.72E+00	1.09E+00	1.67E+00
12	0.934005	2.060150	-0.957952	-	2.06E+00	8.51E-01	1.28E+00
13	0.948901	2.046574	-0.967442	-	1.57E+00	6.63E-01	9.81E-01
14	0.960434	2.036062	-0.974791	-	1.20E+00	5.16E-01	7.54E-01
15	0.969365	2.027922	-0.980481	-	9.21E-01	4.01E-01	5.80E-01
16	0.976279	2.021620	-0.984886	-	7.08E-01	3.12E-01	4.47E-01
17	0.981633	2.016740	-0.988298	-	5.45E-01	2.42E-01	3.45E-01
18	0.985779	2.012962	-0.990939	-	4.21E-01	1.88E-01	2.67E-01
							
69	0.99999996930	2.00000002798	-0.99999998044	Converged	8.948E-07	4.078E-07	5.701E-07

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

1. Convergence criterion (diagonal dominant) + Pivoting

Notes:

Criterion is sufficient (will happen) but not necessary (can happen) for convergence (that is, if the coefficients do not meet the criterion, convergence can still happen).

We can use partial pivoting to rearrange the equations and achieve diagonal dominance.

The following set of equations is not diagonally dominant:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ 1 & -1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}$$

We can apply partial pivoting:

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 2 \\ 1 \\ 3 \end{Bmatrix}$$

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

1. Convergence criterion (diagonal dominant) + Pivoting

Example 3.8b - check convergence criterion, pivot if necessary, solve using Gauss Seidel

$$8x_1 + 3x_2 + x_3 = 12$$

$$-6x_1 + 0x_2 + 7x_3 = 1$$

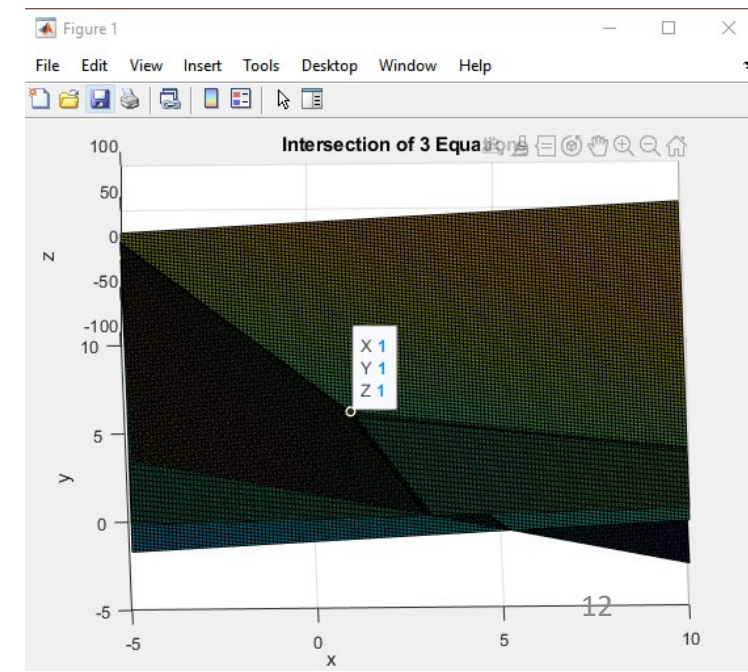
$$2x_1 + 4x_2 - x_3 = 5$$

$$\begin{bmatrix} 8 & 3 & 1 \\ 2 & 4 & -1 \\ -6 & 0 & 7 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 12 \\ 5 \\ 1 \end{Bmatrix}$$

Then convergence is guaranteed (diagonal dominant matrix).

$$|a_{ii}| > \sum_{j=1; j \neq i}^n |a_{ij}|$$

Graphical Method



3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

1. Convergence criterion (diagonal dominant) + Pivoting (ex. 3.8b)

$$\begin{bmatrix} 8 & 3 & 1 \\ 2 & 4 & -1 \\ -6 & 0 & 7 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 12 \\ 5 \\ 1 \end{Bmatrix}$$

$$\text{Row\#1 } x_1 = \frac{1}{8} [12 - 3x_2 - 1x_3]$$

$$\text{Row\#2 } x_2 = \frac{1}{4} [5 - 2x_1 + 1x_3]$$

$$\text{Row\#3 } x_3 = \frac{1}{7} [1 + 6x_1 - 0x_2]$$

Iteration #	x1	x2	x3		Convergence ->		
					εa1	εa2	εa3
0	0.000000000000	0.000000000000	0.000000000000				1.00E-06
1	1.500000000000	0.500000000000	1.42857142857	-	1.000E+02	1.000E+02	1.000E+02
2	1.13392857143	1.04017857143	1.11479591837	-	3.228E+01	5.193E+01	2.815E+01
3	0.97058354592	1.04340720663	0.97478589650	-	1.683E+01	3.094E-01	1.436E+01
4	0.98687406045	1.00025944390	0.98874919467	-	1.651E+00	4.314E+00	1.412E+00
5	1.00130905920	0.99653276907	1.00112205075	-	1.442E+00	3.740E-01	1.236E+00
6	1.00115995526	0.99970053506	1.00099424736	-	1.489E-02	3.169E-01	1.277E-02
7	0.99998801843	1.00025455262	0.99998973009	-	1.172E-01	5.539E-02	1.005E-01
8	0.99990582651	1.00004451927	0.99991927986	-	8.220E-03	2.100E-02	7.046E-03
9	0.99999339529	0.99998312232	0.99999433882	-	8.757E-03	6.140E-03	7.506E-03
10	1.00000703678	0.99999506632	1.00000603152	-	1.364E-03	1.194E-03	1.169E-03
11	1.00000109619	1.00000095979	1.00000093959	-	5.941E-04	5.893E-04	5.092E-04
12	0.99999952263	1.00000047358	0.99999959083	-	1.574E-04	4.862E-05	1.349E-04
13	0.99999987355	0.9999996093	0.99999989162	-	3.509E-05	5.127E-05	3.008E-05
14	1.00000002820	0.99999995880	1.00000002417	-	1.546E-05	2.125E-07	1.326E-05
15	1.00000001243	0.99999999983	1.00000001065	-	1.577E-06	4.102E-06	1.352E-06
16	0.99999999873	1.00000000330	0.99999999891	-	1.369E-06	3.467E-07	1.174E-06
17	0.99999999890	1.00000000028	0.99999999906	Converged	1.670E-08	3.018E-07	1.431E-08

* Initial guesses set to zero. If we had picked 1.0, we would have guessed the correct answer.

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

1. Convergence criterion (diagonal dominant)

For Gauss Seidel, the solution of $[A]\{x\}=\{b\}$, convergence will occur only if $[A]$ is diagonally dominant. [section 11.2 in Chapra]

$$|a_{ii}| > \sum_{j=1; j \neq i}^n |a_{ij}|$$

Consider two equations:

$$u: 11x_1 + 13x_2 = 286$$

$$v: 11x_1 - 9x_2 = 99$$

If 'u' is first, our matrix is not diagonal dominant and the solution diverges.

If we pivot to make 'v' the first equation, we have a diagonal dominant matrix and converges in 54 iterations.

u in first row
[diverges]

Iteration #	x1	x2
0	0.000000	0.000000
1	26.000000	20.777778
2	1.444444	-9.234568
3	36.913580	34.116598
4	-14.319616	-28.501753
5	59.683890	61.946976
6	-47.210063	-68.701188
7	107.192313	120.012827
8	-115.833341	-152.574083
9	206.314826	241.162565
10	-259.010304	-327.568149
11	413.125995	493.931771
12	-557.737548	-692.679225
13	844.620902	1021.314436
14	-1181.007970	-1454.454186
15	1744.900401	2121.656046
16	-2481.411691	-3043.836511
17	3623.261331	4417.430516
18	-5194.599701	-6359.955190
19	7542.310679	9207.379718
20	-10855.448758	-13278.770704

v in first row
[converges]

Iteration #	x1	x2
0	0.000000	0.000000
1	9.000000	14.384615
2	20.769231	4.426036
3	12.621302	11.320437
4	18.262176	6.547390
5	14.356955	9.851807
6	17.060569	7.564134
7	15.188837	9.147908
8	16.484652	8.051449
9	15.587549	8.810536
10	16.208620	8.285014
11	15.778648	8.648837
12	16.076321	8.396959
13	15.870239	8.571336
14	16.012911	8.450614
15	15.914138	8.534191
16	15.982520	8.476330
17	15.935179	8.516387
18	15.967953	8.488655
19	15.945263	8.507854
20	15.960972	8.494562



54	15.954545	8.500000
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3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

1. Convergence criterion (diagonal dominant)

Cobweb plots

$$u: 11x_1 + 13x_2 = 286$$

$$v: 11x_1 - 9x_2 = 99$$

u in first row
[diverges]

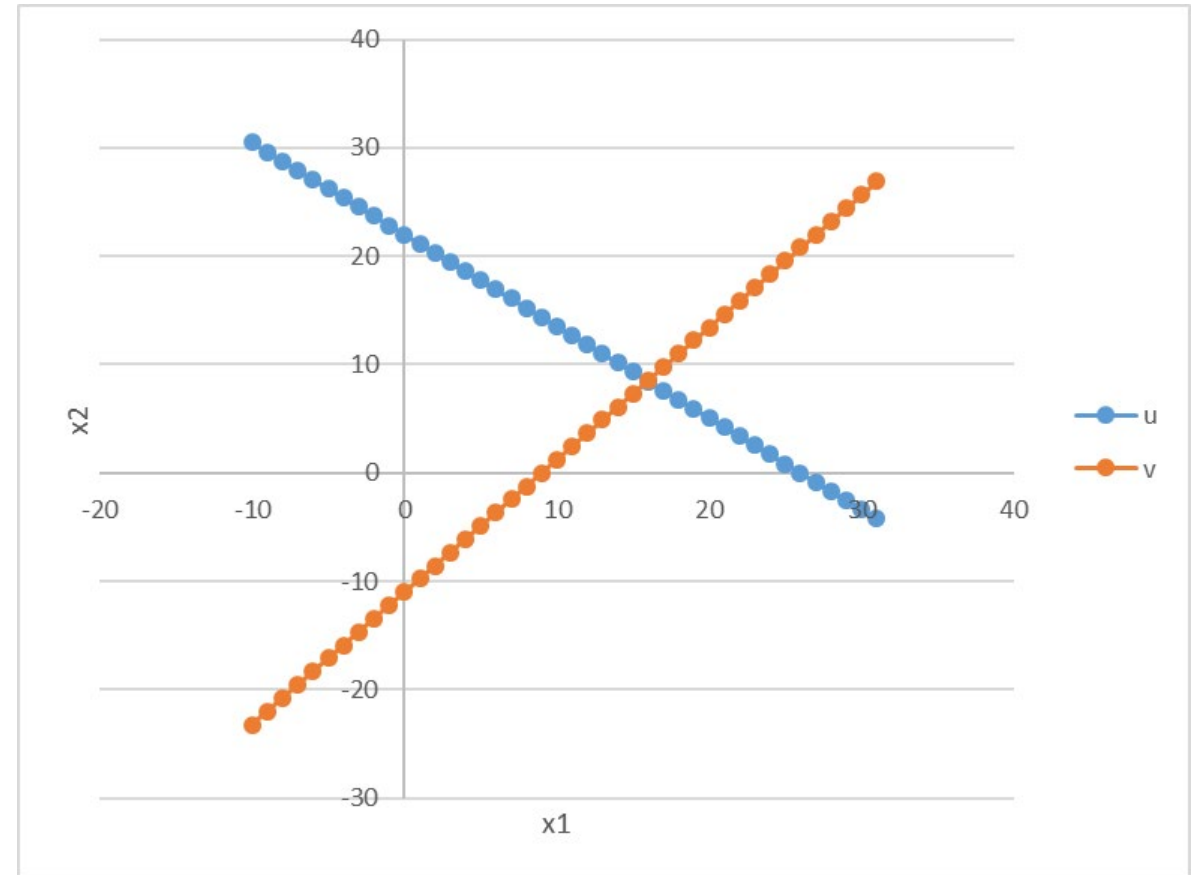
Iteration #	x1	x2
0	0.000000	0.000000
1	26.000000	20.777778
2	1.444444	-9.234568
3	36.913580	34.116598
4	-14.319616	-28.501753
5	59.683890	61.946976
6	-47.210063	-68.701188
7	107.192313	120.012827
8	-115.833341	-152.574083
9	206.314826	241.162565
10	-259.010304	-327.568149
11	413.125995	493.931771
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18	-5194.599701	-6359.955190
19	7542.310679	9207.379718
20	-10855.448758	-13278.770704

v in first row
[converges]

Iteration #	x1	x2
0	0.000000	0.000000
1	9.000000	14.384615
2	20.769231	4.426036
3	12.621302	11.320437
4	18.262176	6.547390
5	14.356955	9.851807
6	17.060569	7.564134
7	15.188837	9.147908
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18	15.967953	8.488655
19	15.945263	8.507854
20	15.960972	8.494562



54	15.954545	8.500000
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3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

Example 3.8c: Check convergence criterion, pivot if necessary, solve using Gauss Seidel

$$\begin{aligned} -x_1 + 3x_2 + 5x_3 &= 7 \\ -2x_1 + 4x_2 - 5x_3 &= -3 \\ 2x_2 - x_3 &= 1 \end{aligned}$$

$$\begin{aligned} -2x_1 + 4x_2 - 5x_3 &= -3 \\ 0x_1 + 2x_2 - 1x_3 &= 1 \\ -x_1 + 3x_2 + 5x_3 &= 7 \end{aligned}$$

<- After pivoting

$$\begin{bmatrix} -2 & 4 & -5 \\ 0 & 2 & -1 \\ -1 & 3 & 5 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -3 \\ 1 \\ 7 \end{Bmatrix}$$

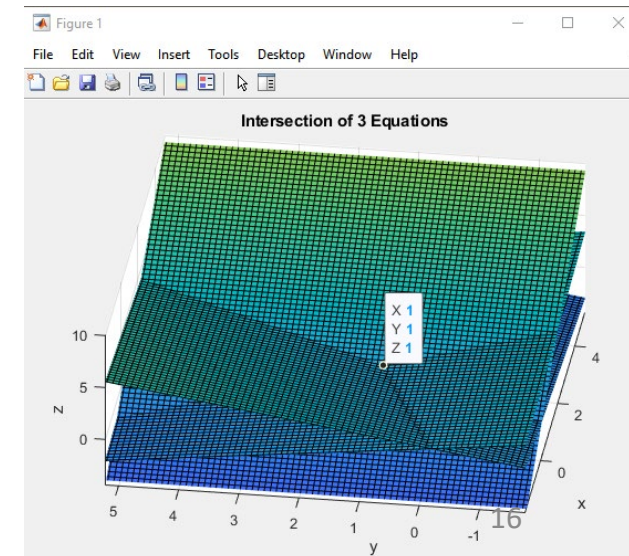
Recall $|a_{ii}| > \sum_{j=1; j \neq i}^n |a_{ij}|$
for convergence.

These equations fail in all 3 rows, therefore
we need to apply pivoting.

Row #1 does not meet the criterion.

Let's see if the problem converges
using Gauss Seidel.

Graphical Method



3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

Example 3.8c: Check convergence criterion, pivot if necessary, solve using Gauss Seidel

Converges in one step, if we guess the correct solution.

$$\begin{bmatrix} -2 & 4 & -5 \\ 0 & 2 & -1 \\ -1 & 3 & 5 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -3 \\ 1 \\ 7 \end{Bmatrix}$$

				Convergence ->			1.00E-06	Original Equations (before pivoting)			
Iteration #	x1	x2	x3		εa1	εa2	εa3		x1	x2	x3
0	1.000000	1.000000	1.000000						1.00	1.00	1.00
1	1.000000	1.000000	1.000000	Converged	0.00E+00	0.00E+00	0.00E+00				
2	1.000000	1.000000	1.000000	Converged	0.00E+00	0.00E+00	0.00E+00				
								CHECK LHS			RHS
								Row1	7.00		7.00
								Row2	-3.00		-3.00
								Row3	1.00		1.00

Row#1 $x_1 = \frac{1}{-2} [-3 - 4x_2 + 5x_3]$

Row#2 $x_2 = \frac{1}{2} [1 - 0x_1 + 1x_3]$

Row#3 $x_3 = \frac{1}{5} [7 + 1x_1 - 3x_2]$

Diverges even if guesses are close to the actual solution

Iteration #	x1	x2	x3		εa1	εa2	εa3
0	1.010000	1.010000	1.010000				
1	0.995000	1.005000	0.996000	-	1.51E+00	4.98E-01	1.41E+00
2	1.020000	0.998000	1.005200	-	2.45E+00	7.01E-01	9.15E-01
3	0.983000	1.002600	0.995040	-	3.76E+00	4.59E-01	1.02E+00
4	1.017600	0.997520	1.005008	-	3.40E+00	5.09E-01	9.92E-01
5	0.982520	1.002504	0.995002	-	3.57E+00	4.97E-01	1.01E+00
6	1.017504	0.997501	1.005000	-	3.44E+00	5.02E-01	9.95E-01
7	0.982501	1.002500	0.995000	-	3.56E+00	4.99E-01	1.01E+00
8	1.017500	0.997500	1.005000	-	3.44E+00	5.01E-01	9.95E-01
9	0.982500	1.002500	0.995000	-	3.56E+00	4.99E-01	1.01E+00
10	1.017500	0.997500	1.005000	-	3.44E+00	5.01E-01	9.95E-01
11	0.982500	1.002500	0.995000	-	3.56E+00	4.99E-01	1.01E+00
12	1.017500	0.997500	1.005000	-	3.44E+00	5.01E-01	9.95E-01
13	0.982500	1.002500	0.995000	-	3.56E+00	4.99E-01	1.01E+00
14	1.017500	0.997500	1.005000	-	3.44E+00	5.01E-01	9.95E-01
15	0.982500	1.002500	0.995000	-	3.56E+00	4.99E-01	1.01E+00
16	1.017500	0.997500	1.005000	-	3.44E+00	5.01E-01	9.95E-01
1000	1.017500	0.997500	1.005000	-	3.44E+00	5.01E-01	9.95E-01

No convergence up to ~1000 iterations

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

2. Relaxation – this is a method of changing the convergence rate of the iterative solution by including a ‘relaxation’ parameter (ω)

$$[A] \{x\} = \{b\}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

Let's look at Row#1, for Gauss Seidel:

$$x_1^0 = 1.0 \text{ [Initial guesses]}$$

$$x_1^1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2^0 - a_{13}x_3^0]$$

Now add the relaxation parameter (ω) to weight the previous and current values of x_1 .

$$x_1^{1'} = \omega x_1^1 + (1 - \omega)x_1^0$$

As we iterate, this will become:

$$x_n^{new} = \omega x_n^{GS} + (1 - \omega)x_n^{old}$$

or

$$x_n^{i'} = \omega x_n^i + (1 - \omega)x_n^{i-1}$$

n= variable #

i = iteration#

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

2. Relaxation parameter (ω)

$$x_n^{new} = \omega x_n^{GS} + (1 - \omega)x_n^{old}$$

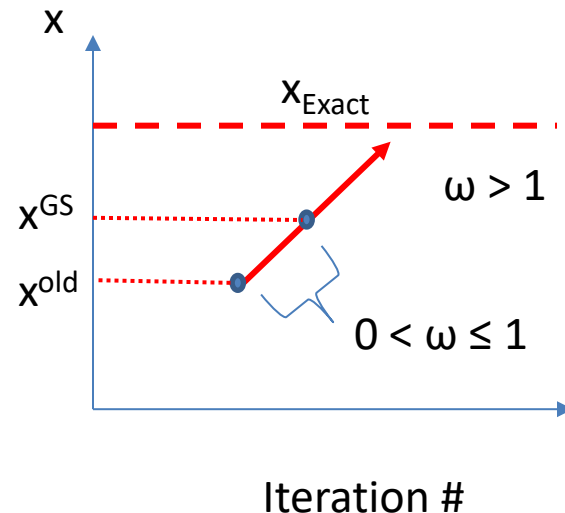
Typically $1 \leq \omega < 2$

Chapra uses the variable λ for the relaxation parameter

$\omega = 1.0$ [this is Gauss Seidel, no relaxation]

$\omega < 1.0$ [called under relaxation, slows convergence and controls divergence]

We can also use relaxation to speed convergence:



Assume new value is found by moving in the direction of x .

For $\omega > 1$, we move closer to x_{exact} , compared to $\omega = 1$.

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

2. Over-relaxation ($\omega > 1$)

$$x_n^{new} = \omega x_n^{GS} + (1 - \omega)x_n^{old}$$

$\omega > 1.0$ [called over-relaxation, speeds convergence]

Can become divergent when $\omega > 2$ (or in some cases when close to 2)

Including the relaxation parameter

$$\text{Row\#1 } x_1^{new} = \frac{\omega}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3] + (1 - \omega)x_1$$

$$\text{Row\#2 } x_2^{new} = \frac{\omega}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3] + (1 - \omega)x_2$$

$$\text{Row\#3 } x_3^{new} = \frac{\omega}{a_{33}} [b_3 - a_{31}x_1 - a_{32}x_2] + (1 - \omega)x_3$$

3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

2. Gauss Seidel with Relaxation (ω) $x_n^{new} = \omega x_n^{GS} + (1 - \omega)x_n^{old}$

Example 3.8d (same equations from ex. 3.7a)

$$2x_1 + x_2 + x_3 = 3$$

$$x_1 + 2x_2 - x_3 = 6$$

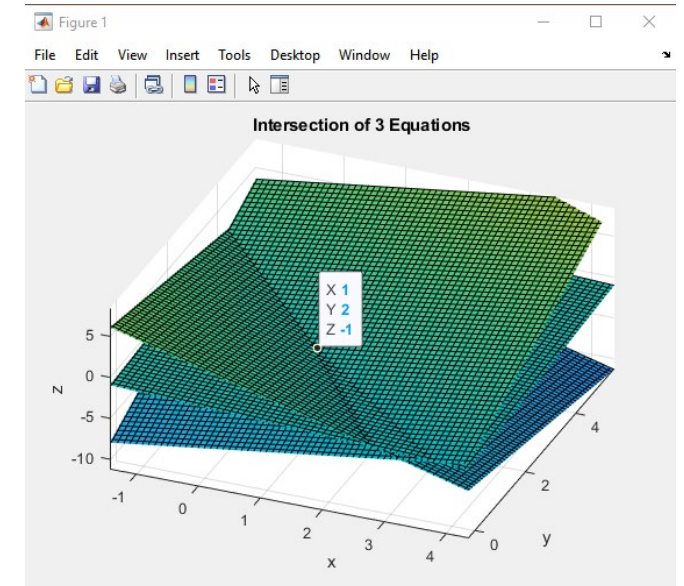
$$x_1 - x_2 + 3x_3 = -4$$

$$x_1^{new} = \frac{\omega}{2}[3 - 1x_2 - 1x_3] + (1 - \omega)x_1$$

$$x_2^{new} = \frac{\omega}{2}[6 - 1x_1 + 1x_3] + (1 - \omega)x_2$$

$$x_3^{new} = \frac{\omega}{3}[-4 - 1x_1 + 1x_2] + (1 - \omega)x_3$$

Graphical Method



3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

2. Gauss Seidel with Relaxation (ω) $x_n^{new} = \omega x_n^{GS} + (1 - \omega)x_n^{old}$

Example 3.8d

$$2x_1 + x_2 + x_3 = 3$$

$$x_1 + 2x_2 - x_3 = 6$$

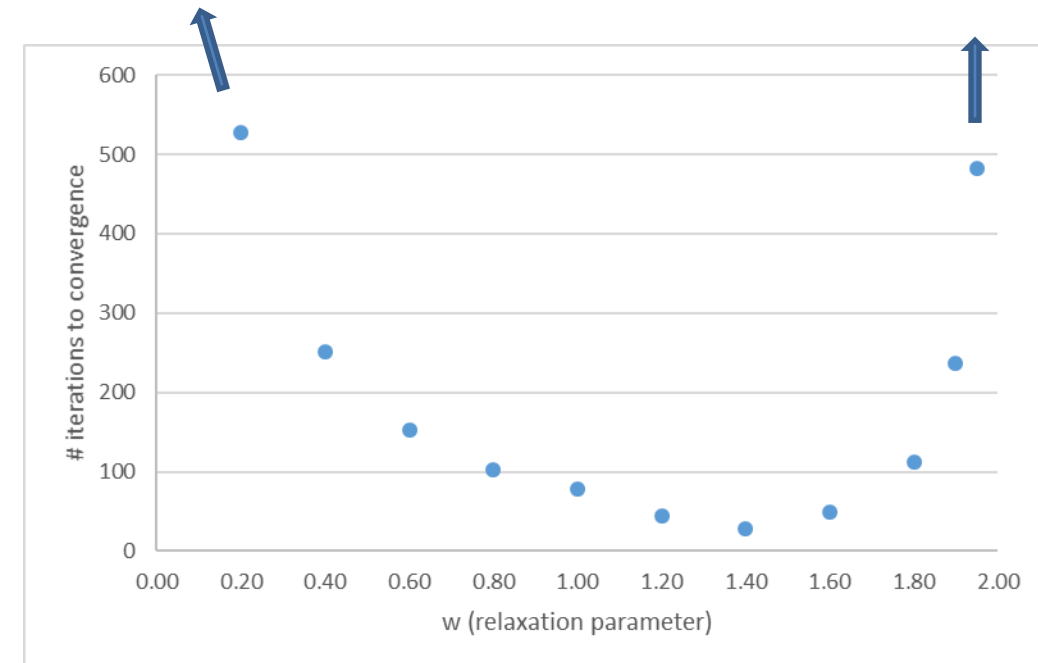
$$x_1 - x_2 + 3x_3 = -4$$

Examples with different ω values.

$$0 < \omega < 2$$

Recall $\omega=1$ is the standard
Gauss Seidel formulation.

w	# iterations
0.20	528
0.40	251
0.60	153
0.80	102
1.00	79
1.20	45
1.40	29
1.60	50
1.80	113
1.95	482
2.00	Diverges



3.8 Systems of Linear Equations [Gauss Seidel w/ Relaxation]

Let's revisit Example 3.8c: Check convergence criterion, pivot if necessary, solve using Gauss Seidel w/ relaxation

$$\begin{bmatrix} -2 & 4 & -5 \\ 0 & 2 & -1 \\ -1 & 3 & 5 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -3 \\ 1 \\ 7 \end{Bmatrix}$$

$$\text{Row\#1 } x_1 = \frac{1}{-2} [-3 - 4x_2 + 5x_3]$$

$$\text{Row\#2 } x_2 = \frac{1}{2} [1 - 0x_1 + 1x_3]$$

$$\text{Row\#3 } x_3 = \frac{1}{5} [7 + 1x_1 - 3x_2]$$

$\omega=1 \rightarrow$ solution diverges (as before)

$\omega>1 \rightarrow$ solution diverges

$\omega<1 \rightarrow$ solution converges

$\omega=0.9$ converges in 36 iterations

$\omega=0.8$ converges in 17 iterations

$\omega=0.7$ converges in 20 iterations

Check in Original Equations			
	x1	x2	x3
sol'n->	1.00	1.00	1.00
CHECK LHS			RHS
Row1	-3.00		-3.00
Row2	1.00		1.00
Row3	7.00		7.00
	Check (abs diff) ->		0.0000